Performance-Based Fault-Tolerant Control Uniformly Accommodating Multiple Levels of Actuation Failures

Ye Cao, Xinjun Wang, Yajing Ma, Jianfu Cao, and Yongduan Song, *Fellow, IEEE*

***Abstract*—Most existing fault-tolerant control schemes either are contingent upon timely and accurate fault identification or need to be redesigned whenever the types/phases of actuator failures change, rendering the control design rather complex. In this work, a dynamic hardware redundancy technique with a performance-based trivial fault detector is proposed to accommodate multiple levels of actuator failures, upon which a unified performance-based fault-tolerant control scheme is developed irrespective of different phases of actuator failures. The proposed fault-tolerant control exhibits several salient features, which include the following: by choosing appropriate performance functions, the controller is developed under a unified control structure against different actuation phases ranging from healthy actuation and moderate failure actuation to post extreme failure actuation; the faulty actuator is replaced with the backup one only in the event of extreme failures, avoiding the unnecessary replacement during moderate actuation failure; and without the need for neural networks or fuzzy systems based approximation, the resultant control scheme is simple in structure and inexpensive in computation.**

***Index Terms*—Actuator failures, fault-tolerant control, performance recovery, prescribed performance.**

I. Introduction

For a safety-critical system, strong fault-tolerant capability against unexpected actuator faults or failures is extremely important [1], [2]. Actuator faults can be broadly classified into "moderate failure" (i.e., the actuator, although losing its effectiveness, is still functional) and "extreme failure" (i.e., the actuator fails to work completely). The past decades have witnessed much progress on fault-tolerant control against moderate failures, such as passive fault-tolerant control scheme [3], information-based fault diagnostic and fault-tolerant control method [4], direct adaptive-state feedback control scheme [5], and so on. These methods have the advantage of low cost and high effectiveness because they compensate for the failure's adverse effects by adjusting the control loop without replacing the faulty actuator. However, if the actuator cannot affected by the controller, especially in the case of extreme failures, the fault-tolerant control approaches described previously are no longer effective. Fortunately, hardware redundancy technique is capable of addressing extreme actuator failures. In [6] and [7], a static hardware redundancy control architecture is employed, where two or more actuators are operated in parallel and the effect of the faulty actuator output can be compensated by the remaining healthy one. The advantage of this static redundancy is that actuation faults are accommodated automatically without the need for fault detection, simplifying the control structure. However, when confronted adverse environment, such as overheating and loss of power supply, all of the actuators may fail at the same time. To deal with such situation, dynamic hardware redundancy control architecture with a fault supervision module and an actuator replacement rule is developed in [8] and [9]. In this case, only one actuator operates on each controller-to-actuator channel at a certain time, and the faulty actuator will be replaced with a healthy one when the severity of the failure triggers the designed supervision conditions. Thus, dynamic hardware redundancy is capable of handling a larger class of failures than static hardware redundancy, but the fault supervision mechanism and switched control architecture make the control design more complicated.

Nowadays, performance-based fault detection methods and performance-guaranteed fault-tolerant control methods gain much attention [10]. Furthermore, fault-tolerant control for strict/pure-feedback systems is of great significance. In [11], [12], and [13], adaptive controllers are proposed to compensate for the adverse effect of actuator failures in strict-feedback systems. To further ensure both transient and steady-state performance, the integration of prescribed performance control approaches are integrated with a fault-tolerant control architecture for strict-feedback systems in [14], [15], and [16]. The performance-based fault-tolerant control methods mentioned previously in [14], [15], and [16] are only able to handle moderate failures. For strict-feedback systems with extreme actuator failure, a switching control strategy and a performance-based fault supervision mechanism are proposed in [17] and [18], where the faulty actuator is switched to the backup one if the tracking error triggers the fault supervision condition. However, the fault-tolerant control methods in [17] and [18] require no delay in detecting the failure and replacing the faulty actuator. Thus, Ouyang and Yan [19] investigate the referred actuator replacement and the performance recovery control after the faulty actuator replacement. To the best of the authors' knowledge, there remain the following two unsolved issues in existing performance-based fault-tolerant control approaches.

1) Most of the existing performance-guaranteed fault-tolerant control methods, such as [14], [15], and [16], are primarily designed for handling moderate failure. Although the methods in [19] can

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cope with extreme failure case, the controllers need to be fully redesigned before and after the replacement of a faulty actuator, which complicates the control design process and limits its practical applicability. Thus, how to design a unified control structure that can deal with different phases of actuator failures and adapt to multiple types of failures has not been solved.

2) In the hardware redundancy architecture, if the performance-based failure detection conditions are met, the faulty actuator will be replaced. However, the performance-guaranteed controllers in [17], [18], and [19] are designed solely for application in free-failure situations. As a result, any failure might activate the performance-based failure detection mechanism, resulting in frequent actuator switching.

To solve the aforementioned issues, we design a performance-based fault-tolerant control method applicable to the entire process of system operation. The originality and main contributions of this article are summarized as follows.

1) A unified control structure is proposed for different phases of actuator failures with which prescribed performance is ensured during moderate failure phase, while performance recovery is achieved within predefined time during the post extreme failure phase.
2) The actuator situations are classified into fault-free, moderate actuator failure, and extreme actuator failure in this article. And the prescribed performance-guaranteed controller is designed to accommodate both free-failure case and moderate failure case, resulting in a lower actuator switching frequency compared to the results in [17], [18], and [19].
3) Although nonaffine systems with uncertainties and disturbances are addressed in the system, the proposed control scheme does not need any approximation method (i.e., neural network or fuzzy system), rendering the resultant fault-tolerant controller simple in structure and inexpensive in computation.

II. Problem Statement

The following nonlinear system in pure-feedback form is considered:

$$\begin{aligned}\dot{x}_i &= f_i(x_1,\ldots,x_i,x_{i+1})+d_i(t),\quad i=1,\ldots,n-1\\ \dot{x}_n &= f_n(x_1,\ldots,x_n,u_a)+d_n(t)\\ y &= x_1\end{aligned}\tag{1}$$

where $x=[x_1,x_2,\ldots,x_n]^T\in\mathbb{R}^n$ is the state vector of the system, $y\in\mathbb{R}$ is the system output, and $u_a\in\mathbb{R}$ is the control input. Functions $f_i:\mathbb{R}^{i+1}\to\mathbb{R}(i=1,\ldots,n)$ are unknown continuously differentiable functions, and $d_i(t):\mathbb{R}_+\to\mathbb{R}(i=1,\ldots,n)$ are unknown piecewise continuous disturbances.

As unanticipated, different types of actuator faults may occur during long-term operation, such as actuator bias faults, actuator partial loss-of-effectiveness faults, actuator lock-in-place faults, and actuator fully loss-of-effectiveness faults, which are defined as follows.

Actuator bias faults model [20]

$$u_a(t)=u(t)+u_d(t)\tag{2}$$

where $u_a(t)$ is the actuator output (plant's actual input), $u(t)$ is the actuator input (our designed control input), and $u_d(t)$ denotes the actuator bias fault, that can be fast time-varying or intermittently abrupt.

Actuator partial/fully loss-of-effectiveness faults model [21]

$$u_a(t)=\rho(t)u(t)\tag{3}$$

where $\rho(t)$ is the "health indicator" reflecting the effectiveness of the actuator with $0\le\rho(t)\le 1$. If there exists a constant $\rho_0>0$ such that $0<\rho_0<\rho(t)\le 1$, the actuator is partial loss-of-effectiveness faults. If the health indicator $\rho(t)=0$, the actuator is fully loss-of-effectiveness faults.

Actuator lock-in-place faults model [6]

$$u_a(t)=\bar{u}\tag{4}$$

where $\bar{u}$ stands for the place of stuck actuator.

The actuator failure models in (2)–(4) can be rewritten uniformly as

$$u_a(t)=\rho(t_\rho,t)u(t)+u_d(t_d,t)\tag{5}$$

where $u_a(t)$ is the actuator output, $u(t)$ is the designed control input, t_ρ denotes the time instant at which the loss of actuator effectiveness fault takes place, and t_d denotes the time instant at which the actuator bias fault or struck fault occurs. Note that the actuator bias $u_d(t_d,t)$ are bounded piecewise continuous functions satisfying $|u_d(t_d,t)|<\bar{u}_d$ and the health indicator $0\le\rho(t_\rho,t)\le 1$.

When the health indicator $0<\rho(\cdot)\le 1$, the model (5) contains bias faults and partial loss-of-effectiveness faults. Since the actuator output $u_a(t)$ still can be influenced by the designed control input $u(t)$, such actuator faults are called "moderate failure" in this article. On the other hand, when the health indicator $\rho(\cdot)=0$, the model (5) contains the lock-in-place faults and fully loss-of-effectiveness faults. In this situation, the actuator output $u_a(t)$ is no longer affected by the designed control scheme $u(t)$, and such actuator faults are called "extreme failure." Thus, the fault model (5) represents multiple types of failures.

This article focuses on the control design for systems with different types/phases of actuator failures, which includes the following aspects:

1) guaranteeing the prescribed performance for actuator in health and moderate fault phases;
2) detecting the occurrence of extreme actuator faults;
3) recovering the system performance from performance violation after the extreme faulty actuator is replaced with a healthy one.

The following assumptions are needed to achieve the control objectives.

Assumption 1: The functions $f_i:\mathbb{R}^{i+1}\to\mathbb{R}, i=1,\ldots,n$ are continuously differentiable, and the partial derivative of $\frac{\partial f_i(x_1,\ldots,x_i,x_{i+1})}{\partial x_{i+1}}$ and $\frac{\partial f_n(x_1,\ldots,x_n,u_a)}{\partial u_a}$ are nonzero, i.e., there exist unknown positive constants b_i such that

$$\left|\frac{\partial f_i(x_1,\ldots,x_i,x_{i+1})}{\partial x_{i+1}}\right|\ge b_i>0, i=1,\ldots,n-1\tag{6}$$

$$\left|\frac{\partial f_n(x_1,\ldots,x_n,u_a)}{\partial u_a}\right|\ge b_n>0\tag{7}$$

for all $(x_1,\ldots,x_n,u_a)\in\mathbb{R}^{n+1}$.

Assumption 2: The signs of $\frac{\partial f_i(x_1,\ldots,x_i,x_{i+1})}{\partial x_{i+1}}$ and $\frac{\partial f_n(x_1,\ldots,x_n,u_a)}{\partial u_a}$ are known. Given the conditions $\frac{\partial f_i(x_1,\ldots,x_i,x_{i+1})}{\partial x_{i+1}}\ne 0$ and $\frac{\partial f_n(x_1,\ldots,x_n,u_a)}{\partial u_a}\ne 0$, we can conclude that each of these partial derivatives is either strictly positive or strictly negative. Without loss the generality, we assume all the signs are positive here.

Assumption 3: The desired trajectory y_d is a continuously differentiable and bounded function, and its first derivative is also bounded.

Assumption 4: The unknown time-varying disturbances $d_i(t)(i=1,\ldots,n)$ are piecewise continuous and bounded, i.e., $|d_i(t)|<D_i$ with D_i as unknown constants.

Remark 1: In Assumption 1, $\frac{\partial f_i(x_1,\ldots,x_i,x_{i+1})}{\partial x_{i+1}}\ne 0$ is to guarantee the controllability of system. Assumption 2 is equivalent to demanding that the control direction is known. It is interesting to note that, except for the sign of $\frac{\partial f_i(x_1,\ldots,x_i,x_{i+1})}{\partial x_{i+1}}$ and the continuous differentiability requirement of $f_i(\cdot)$, no other information on $f_i(\cdot)$ is needed in our

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control design. On the other hand, fault-tolerant control is investigated in the case of actuator failure but no sensor failure.

III. Performance-Based Control Architecture for the Entire Process of System Operation

In order to incorporate system performance considerations into the design of the control architecture, some performance functions are defined in this section to symbolize the desired convergence speed and convergence accuracy.

A. Prescribed Performance Function

Before moving on, tracking error and other error variables for the backstepping technique are defined as follows:

$$z_1 = y - y_r \tag{8}$$

$$z_i = x_i - \alpha_{i-1}, \qquad i = 2, \ldots, n \tag{9}$$

where α_{i-1} is the virtual controller to be designed later.

Now the prescribed performance function $\beta_1(t)$ for the tracking error $z_1(t)$ is defined as

$$\beta_1(t) = (p_{10} - p_{1\infty})\kappa(t) + p_{1\infty} \tag{10}$$

where p_{10} represents the initial value of the performance function satisfying $p_{10} > |z_1(0)|$, $p_{1\infty}$ is chosen based on the steady-state tracking accuracy with $0 < p_{1\infty} < p_{10}$, and $\kappa(t)$ is selected according to the desired tracking error convergence speed. Besides, $\kappa(t)$ is a continuously differentiable function and monotonously decreases from 1 to 0 with $\kappa(0) = 1$ and $\kappa(\infty) = 0$. Many functions such as $e^{-\lambda t}$, $1/(1+t)$, $1/(1+t^2)$, and $1/[2^t(1+t^2)]$ can be served as $\kappa(t)$. Thus, if the tracking error $z_1(t)$ satisfies

$$|z_1(t)| < \beta_1(t) \tag{11}$$

it can be concluded that the tracking error $z_1(t)$ converges to the interval $(-p_{1\infty}, p_{1\infty})$ at a rate consistent with the function $\kappa(t)$.

In addition, the virtual errors $z_i(t)$ for the high-order system (1) have an impact on the state of the tracking error $z_1(t)$. For simplicity of analysis, the performance function for $z_i(t)$ is defined as a positive constant β_i satisfying $\beta_i > |z_i(0)|$. If the virtual errors $z_i(t)$ satisfy

$$|z_i(t)| < \beta_i, \qquad i = 2, \ldots, n \tag{12}$$

they are bounded in the backstepping design.

The construction of the performance functions serves two purposes: on the one hand, guaranteeing the prescribed performance of the tracking error $z_1(t)$ is one of the control objectives even in the case of moderate actuator failure; on the other hand, these performance functions are also regarded as monitoring functions that detect the extreme actuator failure.

B. Architecture of Fault-Tolerant Control

In this article, the actuation redundancy architecture is adopted to cope with extreme actuator failure and some healthy actuators are available as backup in this article as shown in Fig. 1. The extreme fault supervision module consists of a series of performance functions that is able to detect the extreme failure. If the actuator extreme failure results in a performance degradation of the system that exceeds the tolerable range, the supervisory mechanism would be triggered. Then, the faulty actuator will be replaced with a healthy one, and the controller is reconfigured adaptively.

Suppose that $[t_0, t_1]$ is a period of system operation, which is divided into health and moderate failure phase $[t_0, t_{ef0})$, extreme failure occurrence phase $[t_{ef0}, t_{r0})$, and post extreme failure phase $[t_{r0}, t_1)$. t_{ef0}, t_{d0}, and t_{r0} are defined as the time instants for extreme failure occurrence, extreme failure detection, and actuator replacement, respectively. The response of the tracking error $z_1(t)$ at different actuator failure phases is shown in Fig. 2.

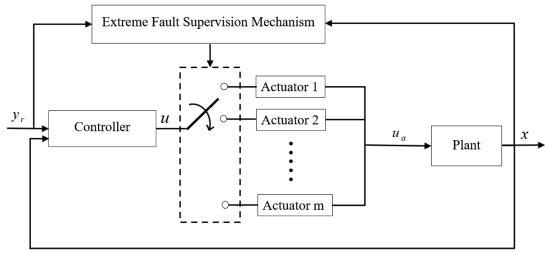


Fig. 1. Architecture of the fault-tolerant system.

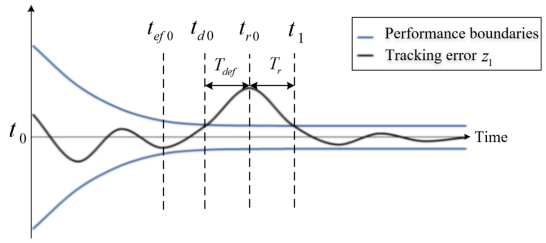


Fig. 2. Response of tracking error at different phases of the actuator failure.

1) During health and moderate failure phase $[t_0, t_{ef0})$, the actuator output $u_a(t)$ is still affected by the control input $u(t)$, thus the desired performance satisfaction (11) can be achieved by designing appropriate fault-tolerant controller.
2) During extreme failure occurrence phase $[t_{ef0}, t_{r0})$, no input is imposed on the plant ($u_a = 0$) or the faulty actuator is stuck on the plant ($u_a = \bar{u}$), indicating that the control loop can no longer compensate for the performance deterioration. Therefore, the primary tasks during this phase are to detect the extreme failures and replace the faulty actuator with a healthy one. In this article, the performance-based extreme fault supervision mechanism is designed as

$$t_{d0} = inf\left\{t : |z_1(t)| \geq \beta_1(t) \text{ or } |z_i(t)| \geq \beta_i, i = 2, \ldots, n\right\}. \tag{13}$$

Since the actuator cannot be replaced and activated immediately, the deferred replacement time T_{def} is allowed, where the healthy actuator is actually activated at time t_{r0}.
3) During the post extreme failure phase $[t_{r0}, t_1)$, the healthy actuator is already available. The control objective is to recover the error $z_1(t)$ from performance violation to performance satisfaction within the prescribed time T_r. After time t_1, the system enters the next period.

The response curves for the virtual errors $z_i(t)$ are similar to these for the error $z_1(t)$, except that the time-varying performance boundaries should be replaced by constant boundaries. The analysis of virtual errors $z_i(t)$ is omitted here.

Remark 2: The extreme actuator failure supervision mechanism (13) is effective on the condition that the errors satisfy the prescribed performance in health and moderate failure phases. As a result, these fault detection mechanisms should be utilized in combination with the performance guaranteed fault-tolerant control strategy described as follows.

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TABLE I
EXAMPLES OF CONSTRAINING FUNCTION

Case	$F(\xi_i), \xi_i \in (-1,1)$	$\frac{\partial F(\xi_i)}{\partial \xi_i}$	$F^{-1}(\varepsilon_i)$
1	$\ln\left(\frac{1+\xi_i}{1-\xi_i}\right)$	$\frac{2}{1-\xi_i^2}$	$\frac{e^{\varepsilon_i}-1}{e^{\varepsilon_i}+1}$
2	$\operatorname{arctanh}(\xi_i)$	$\frac{2}{1-\xi_i^2}$	$\frac{e^{\varepsilon_i}-1}{e^{\varepsilon_i}+1}$
3	$\tan\left(\frac{\pi}{2}\xi_i\right)$	$\frac{\pi}{2\cos^2(\frac{\pi}{2}\xi_i)}$	$\frac{2}{\pi}\arctan(\varepsilon_i)$
4	$\frac{2\xi_i}{(1+\xi_i)(1-\xi_i)}$	$\frac{2(1+\xi_i^2)}{(1-\xi_i^2)^2}$	$\frac{\sqrt{\varepsilon_i^2+1}-1}{\varepsilon_i}$

IV. PERFORMANCE GUARANTEED FAULT-TOLERANT CONTROL FOR MODERATE ACTUATOR FAILURE

In this section, the objective is to design a performance guaranteed fault-tolerant controller such that the tracking error $z_1(t)$ satisfies the performance requirement (11) even in the case of moderate actuator failure $0 < \rho_0 \le \rho(\cdot) \le 1$ with ρ_0 being an unknown positive constant.

A. Error-Dependent Constraining Function

By defining new variables

$$\xi_i(t) = \frac{z_i(t)}{\beta_i(t)},\ i = 1, \ldots, n \tag{14}$$

the desired performance satisfaction problem in (11) and (12) is equivalent to the new variables satisfying

$$|\xi_i(t)| < 1. \tag{15}$$

Thus, in order to ensure the prescribed performance, a constraining function $F(\xi_i) : (-1,1) \to \mathbb{R}$ is constructed with the following properties:

1) $F(\xi_i)$ is a continuously differentiable function with $\xi_i \in (-1,1)$;
2) $F(\xi_i)$ monotonically increases with respect to ξ_i;
3) $F(\xi_i) = 0$ if and only if $\xi_i = 0$;
4) when the variable $\xi_i \to 1$ or $\xi_i \to -1$, the constraining function $F(\xi_i) \to \pm\infty$.

Some examples of constraining functions are shown in Table I, where $\ln(\star)$ denotes the natural logarithm of $\star$ and $\operatorname{arctanh}(\star)$ represents hyperbolic arctangent of $\star$. Based on these properties of $F(\xi_i)$, the following lemma is established.

Lemma 1: For any initial condition satisfying $|\xi_i(t_0)| < 1$, if the constraining function $F(\xi_i(t))$ is bounded for $t \in [t_0, t_{ef0})$, then $|\xi_i(t)| < 1$ holds for all $t \in [t_0, t_{ef0})$.

Proof: Now we give the proof by contradiction. Suppose that $\xi_i(t_0') = \omega$ for $t_0' \in [t_0, t_{ef0})$, where $\omega \ge 1$ or $\omega \le -1$. Since $|\xi_i(t_0)| < 1$ and $\xi_i(t)$ is a continuous function, according to the intermediate value theorem as in [22], there exists a time instant $t_0 \le \bar{t}_0 \le t_0'$ such that $\xi_i(\bar{t}_0) = 1$ or $\xi_i(\bar{t}_0) = -1$. As a result $F(\xi_i(\bar{t}_0)) = \infty$, which leads to a contradiction for the boundedness of $F(\xi_i(t))$ for $t \in [t_0, t_{ef0})$. Thus, the assumption is incorrect and $|\xi_i(t)| < 1$ holds for $t \in [t_0, t_{ef0})$. This completes the proof. □

According to Lemma 1, it is interesting to note that the prescribed performance $|z_i(t)| < \beta_i(t)$ is ensured as long as the constraining function $F(\xi_i(t))$ is bounded. Thus, we focus on ensuring the boundedness of $F(\xi_i(t))$ in the following control design.

B. Control Scheme and Stability Analysis

The backstepping technique is utilized here, where the actual feedback control is recursively derived at the final step. The control design process is summarized as follows.

Step 1: Select p_{10} to be greater than the initial value of error, i.e., $p_{10} > |x_1(t_0) - y_r(t_0)|$, $p_{1\infty}$ as the maximum allowable size of the tracking error at steady-state phase, and $\kappa(t)$ as the desired error convergence rate. Then, the performance function $\beta_1(t)$ as defined in (10) is available for control design. Besides, choose a constraining function $F(\cdot)$ from Table I.

Design the first virtual control signal as

$$\alpha_1(x_1, t) = -k_1 F(\xi_1(t)) = -k_1 F\left(\frac{x_1(t) - y_r(t)}{\beta_1(t)}\right) \tag{16}$$

where k_1 is a positive constant.

Step i ($i = 2, \ldots, n-1$): Select positive constants as the performance function β_i in (12) satisfying $\beta_i > |x_i(0) - \alpha_{i-1}(0)|$.

Design the virtual control signal as

$$\alpha_i(\bar{x}_i, t) = -k_i F(\xi_i(t)) = -k_i F\left(\frac{x_i(t) - \alpha_{i-1}(t)}{\beta_i}\right) \tag{17}$$

where k_i is a positive constant.

Step n: Finally, design the actual control signal as

$$u(\bar{x}_n, t) = -k_n F(\xi_n(t)) = -k_n F\left(\frac{x_n(t) - \alpha_{n-1}(t)}{\beta_n}\right) \tag{18}$$

with k_n being a positive constant and the performance bound satisfying $\beta_n > |x_n(0) - \alpha_{n-1}(0)|$.

Now the results related to the fault-tolerant control (18) are summarized in the following theorem.

Theorem 1: Consider the uncertain nonlinear system (1) subject to moderate failure (5) with $0 < \rho_0 \le \rho(\cdot) \le 1$. Suppose that Assumptions 1–4 hold, if the control design in (16)–(18) is applied, the following objectives are achieved.

1) The closed-loop system is stable and all the internal signals including the control input u are uniformly bounded.
2) The tracking error satisfies the prescribed performance $|z_1(t)| < \beta_1(t)$, where $z_1(t)$ converges to the interval $(-p_{1\infty}, p_{1\infty})$ at a rate consistent with $\kappa(t)$.

Proof: Based on the definition of new variable ξ_i in (14) and error variable z_i in (9), the derivative of $\xi_i(t)$ can be obtained from the system model (1) that

$$\begin{cases} \dot{\xi}_1 = \frac{1}{\beta_1}[f_1(\beta_1\xi_1 + y_r, \beta_2\xi_2 + \alpha_1) + d_1 - \dot{y}_r - \dot{\beta}_1\xi_1] \\ \quad = h_1(t, \xi_1, \xi_2) \\ \dot{\xi}_i = \frac{1}{\beta_i}[f_i(\beta_1\xi_1 + y_r, \ldots, \beta_{i+1}\xi_{i+1} + \alpha_i) + d_i \\ \quad -\frac{\partial \alpha_{i-1}}{\partial \xi_{i-1}} h_{i-1}(t, \xi_1, \ldots, \xi_i) - \dot{\beta}_i\xi_i] \\ \quad = h_i(t, \xi_1, \ldots, \xi_{i+1}) \\ \dot{\xi}_n = \frac{1}{\beta_n}[f_n(\beta_1\xi_1 + y_r, \ldots, \beta_n\xi_n + \alpha_{n-1}, \rho u(\xi_n) + u_d) \\ \quad +d_n - \frac{\partial \alpha_{n-1}}{\partial \xi_{n-1}} h_{n-1}(t, \xi_1, \ldots, \xi_n) - \dot{\beta}_n\xi_n] \\ \quad = h_n(t, \xi_1, \ldots, \xi_n). \end{cases} \tag{19}$$

Since the control inputs are functions with respect to ξ_i, the closed-loop system can be rewritten in a compact form as

$$\dot{\xi} = h(t, \xi) = \begin{bmatrix} h_1(t, \xi_1, \xi_2) \\ \vdots \\ h_n(t, \xi_1, \ldots, \xi_n) \end{bmatrix} \tag{20}$$

with the error vector $\xi = [\xi_1, \ldots, \xi_n]^T$.

Part A: We aim to show the existence and uniqueness of a maximal solution of the close-loop system (20).

Define the open set $\Omega = (-1,1) \times \cdots \times (-1,1)$. Since the performance functions are selected satisfying $\beta_1(0) > |x_1(0) - y_r(0)|$ and $\beta_i > |x_i(0) - \alpha_{i-1}(0)|, i = 2, \ldots, n$, thus the initial condition

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$|\xi_i(0)| < 1 (i = 1, \ldots, n)$ holds, which implies that $\xi(0) \in \Omega$. In addition, function $h(t, \xi)$ is piecewise continuous on t and locally Lipschitz on ξ over the open set Ω, owning to the fact that the nonlinearities $f_i, i = 1, \ldots, n$ and the desired tracking trajectory y_r are continuously differentiable functions, the designed control signals α_i and u are smooth over the set Ω, the disturbance $d_i(t)$, the health indictor ρ, and u_d is piecewise continuous on t. With the aid of the existence and uniqueness theorem for ordinary differential equation (see [23,Th. 3.1]), there exists a maximum time interval $\tau_{\max}$ such that the maximal solution $\xi(t) \in \Omega$ for system (20) on the time interval $[t_0, \tau_{\max})$ is unique.

Part B: We aim to prove that $\xi_i(t)$ is uniformly bounded away from the boundary, i.e., $\exists \bar{\xi}_i > 0$ and $\underline{\xi}_i > 0$

$$s.t. \ -1 < \underline{\xi}_i \leq \xi_i(t) \leq \bar{\xi}_i < 1 \ \forall t \in [t_0, \tau_{\max}). \tag{21}$$

Note that our proposed approximation-free and performance-guaranteed controller is inspired by [24]. The proof process from step 1 to step $n-1$ of the backstepping procedure is similar to those in [24], which has been briefly written. But the proof related to the faulty actuator in step n is presented in detail. For simplicity of analysis, we define $\varepsilon_i = F(\xi_i) (i = 1, \ldots, n)$.

Step $i (i = 1, \ldots, n-1)$: Define the positive definite Lyapunov function candidate $V_i = \frac{1}{2}\varepsilon_i^2$. By using the mean value theorem and the system model (19), the derivative of $\frac{1}{2}\varepsilon_i^2$ is

$$\dot{V}_i = \frac{\varepsilon_i}{\beta_i}\frac{\partial F(\xi_i)}{\partial \xi_i}\Big[f_i(\beta_1\xi_1 + y_r, \ldots, \beta_{i+1}\xi_{i+1}) + d_i - \dot{\alpha}_{i-1} + \alpha_i \frac{\partial f_i(\beta_1\xi_1 + y_r, \ldots, \chi_i)}{\partial \chi_i}\Big|_{\chi_i = \chi_i^*} - \dot{\beta}_i \xi_i\Big] \tag{22}$$

with $\chi_i^* = \lambda(\beta_{i+1}\xi_{i+1} + \alpha_i) + (1-\lambda)\beta_{i+1}\xi_{i+1}$, $\lambda \in (0,1)$, $\dot{\alpha}_0 = \dot{y}_r$, $\dot{\beta}_1 \neq 0$ and $\dot{\beta}_k = 0 (k = 2, \ldots, n-1)$. And the derivative of virtual controller is

$$\dot{\alpha}_{i-1} = \frac{\partial \alpha_{i-1}}{\partial \xi_{i-1}} h_{i-1}(t, \xi_1, \ldots, \xi_i) = -\frac{\partial F(\xi_{i-1})}{\partial \xi_{i-1}}\frac{k_{i-1}}{\beta_{i-1}} \Big[f_{i-1}(\beta_1\xi_1 + y_r, \ldots, \beta_i\xi_i + \alpha_{i-1}) + d_{i-1} - \dot{\alpha}_{i-2} - \dot{\beta}_{i-1}\xi_{i-1}\Big]. \tag{23}$$

Since f_{i-1} is continuous on the set Ω, the variable ξ_{i-1} is bounded on the set Ω, and the boundedness of $\dot{\alpha}_{i-2}$ can be proven in step $i-1$ of backstepping procedure, thus the derivative of virtual controller $\dot{\alpha}_{i-1}$ is also bounded on the set Ω. Hence, there exists a positive constant $\overline{M}_i$ such that $|f_i(\beta_1\xi_1 + y_r, \beta_2\xi_2 + \alpha_1, \ldots, \beta_{i+1}\xi_{i+1}) + d_i - \dot{\alpha}_{i-1} - \dot{\beta}_i\xi_i| \leq \overline{M}_i$ for $\xi \in \Omega$. By substituting the virtual controller (17) into (22), we have

$$\dot{V}_i \leq \frac{1}{\beta_i}\frac{\partial F(\xi_i)}{\partial \xi_i}\left[\overline{M}_i|\epsilon_i| - k_i b_i \varepsilon_i^2\right] \tag{24}$$

with $\frac{\partial f_i(\beta_1\xi_1 + y_r, \ldots, \chi_i)}{\partial \chi_i}\Big|_{\chi_i = \chi_i^*} \geq b_i$. As a result, it is obtained that

$$|\epsilon_i(t)| \leq \bar{\varepsilon}_i = \max\left\{\varepsilon_i(0), \frac{\overline{M}_i}{k_i b_i}\right\}. \tag{25}$$

Therefore, it follows from $\varepsilon_i = F(\xi_i)$ that:

$$-1 < F^{-1}(-\bar{\varepsilon}_i) = \underline{\xi}_i \leq \xi_i(t) \leq \bar{\xi}_i = F^{-1}(\bar{\varepsilon}_i) < 1 \tag{26}$$

for all $t \in [t_0, \tau_{\max})$.

Step n:: By differentiating ε_n with respect to time and employing (19), we have

$$\dot{\varepsilon}_n = \frac{\partial F(\xi_n)}{\partial \xi_n}\dot{\xi}_n = \frac{\partial F(\xi_n)}{\partial \xi_n}\frac{1}{\beta_n}[f_n(\beta_1\xi_1 + y_r, \ldots, \rho u(\xi_n) + u_d) + d_n - \dot{\alpha}_{n-1} - \dot{\beta}_n\xi_n] \tag{27}$$

where $0 < \rho < 1$ is the actuator health indicator and u_d is the actuator bias caused by actuator failure. By using the mean value theorem, the continuous function

$$\begin{aligned} &f_n(\beta_1\xi_1 + y_r, \ldots, \beta_n\xi_n + \alpha_{n-1}, \rho u(\xi_n) + u_d) \\ &= f_n(\beta_1\xi_1 + y_r, \ldots, \beta_n\xi_n + \alpha_{n-1}, u_d) \\ &\quad + \rho u(\xi_n)\frac{\partial f_n(\beta_1\xi_1 + y_r, \ldots, \chi_n)}{\partial \chi_n}\Big|_{\chi_n = \chi_n^*} \end{aligned} \tag{28}$$

where $\chi_n^* = \lambda u_d + (1-\lambda)[\rho u(\xi_n) + u_d]$ with $\lambda \in (0,1)$. By differentiating the Lyapunov function $V_n = \frac{1}{2}\varepsilon_n^2$ and combining (19) and (28), we have

$$\dot{V}_n = \frac{\varepsilon_n}{\beta_n}\frac{\partial F(\xi_n)}{\partial \xi_n}\Big[f_n(\beta_1\xi_1 + y_r, \ldots, \beta_n\xi_n + \alpha_{n-1}, u_d) + d_n - \dot{\beta}_n\xi_n - \dot{\alpha}_{n-1} + \rho u(\xi_n)\frac{\partial f_n(\beta_1\xi_1 + y_r, \ldots, \chi_n)}{\partial \chi_n}\Big|_{\chi_n = \chi_n^*}\Big]. \tag{29}$$

Besides, there exists a positive constant $\overline{M}_n$ such that $|f_n(\beta_1\xi_1 + y_r, \ldots, \beta_n\xi_n + \alpha_{n-1}, u_d) + d_n - \dot{\alpha}_{n-1} - \dot{\beta}_n\xi_n| \leq \overline{M}_n$ for all $\xi \in \Omega$, due to the fact that $f_n(\cdot)$ is a continuous function, the actuator bias satisfies $|u_d(t_d, t)| < \bar{u}_d$, the disturbance $|d_n| < D_n$, ξ_i and $\dot{\alpha}_i (i = 1, \ldots, n-1)$ are bounded for all $\xi \in \Omega$. Therefore, we have

$$\dot{V}_n \leq \frac{1}{\beta_n}\frac{\partial F(\xi_n)}{\partial \xi_n}\left[\overline{M}_n|\epsilon_n| - k_n b_n \rho_0 \varepsilon_n^2\right] \tag{30}$$

with the moderate actuator failure $\rho(\cdot) \geq \rho_0 > 0$. As a result, $\dot{V}_n$ is negative when $|\varepsilon_n| > \frac{\overline{M}_n}{k_n b_n \rho_0}$, which further implies that

$$|\epsilon_n(t)| \leq \bar{\varepsilon}_n = \max\left\{\varepsilon_n(0), \frac{\overline{M}_n}{k_n b_n \rho_0}\right\}. \tag{31}$$

This results in

$$-1 < F^{-1}(-\bar{\varepsilon}_n) = \underline{\xi}_n \leq \xi_n(t) \leq \bar{\xi}_n = F^{-1}(\bar{\varepsilon}_n) < 1 \tag{32}$$

for all $t \in [t_0, \tau_{\max})$. By now, the upper boundary $\bar{\xi}_i > 0$ and lower boundary $\underline{\xi}_i > 0$ for $\xi_i(t)$ with $i = 1, \ldots, n$ have been found, and the proof of part B is complete.

Part C: Now we aim to show that $\tau_{\max} = t_{ef0}$. Let us recall the definition domain of ξ, where $\Omega = (-1, 1) \times \cdots \times (-1, 1)$. Based on the proof in Parts A and B, it can be obtained that the variable $\xi(t)$ is within the set

$$\Omega_\xi' = \left[\underline{\xi}_1, \bar{\xi}_1\right] \times \cdots \times \left[\underline{\xi}_n, \bar{\xi}_n\right] \tag{33}$$

for $t \in [t_0, \tau_{\max}]$ and $\Omega_\xi' \subset \Omega$. If $\tau_{\max} < t_{ef0}$, the solution of the differential equation ξ can be extended to the boundaries of set Ω. In other words, Ω_ξ' should be a compact subset of Ω that contains the whole graph of the solution Ω, which contradicts the fact that $\Omega_\xi' \subset \Omega$. Therefore, it is proven that $\tau_{\max} = t_{ef0}$.

Part D: We aim to prove the conclusion in Theorem 1. The variables ξ_i are bounded and are within the set Ω_ξ' according to the proof in Parts A–C. Thus, the closed-loop system (20) is stable. From (25) and (31), we have that ε_i are bounded. Thus, the virtual controllers $\alpha_i = -k_i\varepsilon_i$ in (16) and (17) and the actual controller $\alpha_n = -k_n\varepsilon_n$

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in (18) are uniformly bounded. The control objective (1) has been proven. On the other hand, since $\xi_1(t) = \frac{z_1(t)}{\beta_1(t)}$ and $-1 < \underline{\xi}_1 \leq \xi_1(t) \leq \bar{\xi}_1 < 1$, thus $|z_1(t)| < \beta_1(t)$. By recalling the definition of $\beta_1(t)$ in Section III-A, it is seen that $z_1(t)$ converges to the interval $(-p_{1\infty}, p_{1\infty})$ at a rate consistent with $\kappa(t)$. The control objective (2) is proven. This completes the proof. □

Remark 3: For the performance guaranteed fault-tolerant controllers in (16)–(18), only the system's initial states and the desired performance are required to design the controller. Although system (2) contains unknown nonlinearities f_i and disturbances d_i, no intelligent methods (such as neural networks and fuzzy systems) are employed to approximate these uncertainties, resulting in a low complexity of control design.

V. Performance Recovery Control After Extreme Actuator Failure

During the post extreme failure phase $[t_{r0}, t_1)$, the faulty actuator has been replaced with the healthy one and the control objective is to recover the tracking error $z_1(t)$ to the original performance boundary $\beta_1(t)$ within a predefined time T_r by using the same control structure as in Section IV.

A. Virtual Performance Function

The performance guaranteed controller (16)–(18) in Section IV is no longer valid, because the initial error $z_1(t_{r0})$ after the extreme faulty actuator replacement is out of the performance boundaries. In order to recover the performance, new virtual performance functions are construct for the post extreme failure phase $[t_{r0}, t_1)$ as

$$\overline{\beta}_i(t) = \begin{cases} (|z_i(t_{r0})| - \beta_i(t_{r0}) + \varepsilon_i)\left(\frac{t_{r0}+T_r-t}{T_r}\right)^2 + \beta_i(t) \\ \qquad \text{if } |z_i(t_{r0})| \geq \beta_i(t_{r0}) \\ \beta_i(t), \quad \text{if } |z_i(t_{r0})| < \beta_i(t_{r0}) \end{cases} \tag{34}$$

where $T_r > 0$ is the recovery time, $\varepsilon_i > 0$ is the positive design parameter and $i = 1, \ldots, n$.

Note that both the desired recovery time T_r and the original performance function $\beta_i(t)$ are included in the virtual performance functions, which have the following properties.

1) If $|z_i(t_{r0})| \geq \beta_i(t_{r0})$, the virtual performance functions at time instant t_{r0} satisfy $\overline{\beta}_i(t_{r0}) = |z_i(t_{r0})| + \varepsilon_i > |z_i(t_{r0})|$. As a result, all errors at $t = t_{r0}$ are within the virtual performance boundaries, i.e., $|z_i(t_{r0})| < \overline{\beta}_i(t_{r0})$;
2) The virtual performance functions satisfy the condition $\lim_{t\to t_1^-} \overline{\beta}_i(t) = \beta_i(t_1)$ and $\lim_{t\to t_1^-} \frac{d}{dt}\overline{\beta}_i(t) = \dot{\beta}_i(t_1)$ at the time instant $t_1 = t_{r0} + T_r$. Thus, even at the beginning of next period t_1, the boundaries $\overline{\beta}_i(t)$ are continuously differentiable.

The original performance boundaries, virtual performance boundaries, and the response of error are shown in Fig. 3. It is concluded that as long as the error constraints

$$|z_i(t)| < \overline{\beta}_i(t) \forall t \in [t_{r0}, t_1) \tag{35}$$

is satisfied, the errors $z_i(t)$ recover from the performance violation to performance satisfaction $|z_i(t)| < \beta_i(t)$ within the predefined time T_r.

B. Reconfigurable Control Design

Now we focus on ensuring that the constraint (35) holds in control design. As long as the initial errors are within the performance boundaries, i.e., $|z_i(t_0)| < \beta_i(t_0)$, the performance guaranteed controller in Theorem 1 is able to guarantee $|z_i(t)| < \beta_i(t)$. Note that the initial errors $z_i(t_{r0})$ are within the virtual performance boundaries, i.e., $|z_i(t_{r0})| < \overline{\beta}_i(t_{r0})$. Thus, by replacing $\beta_i(t)$ with $\overline{\beta}_i(t)$, the virtual performance constraint (35) can be ensured with the same control structure as in Theorem 1. The reconfigurable controller is designed as

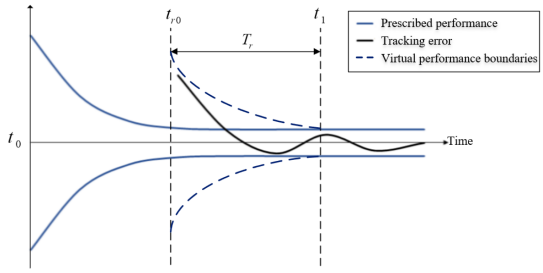


Fig. 3. Response of tracking error and the virtual performance function.

$$\alpha_1(x_1, \bar{\beta}_1, t) = -k_1 F\left(\bar{\xi}_1(t)\right) = -k_1 F\left(\frac{x_1(t) - y_r(t)}{\overline{\beta}_1(t)}\right) \tag{36}$$

$$\alpha_i(\bar{x}_i, \bar{\beta}_i, t) = -k_i F\left(\bar{\xi}_i(t)\right) = -k_i F\left(\frac{x_i(t) - \alpha_{i-1}(t)}{\overline{\beta}_i(t)}\right) \tag{37}$$

$$u(\bar{x}_n, \bar{\beta}_n, t) = -k_n F\left(\bar{\xi}_n(t)\right) = -k_n F\left(\frac{x_n(t) - \alpha_{n-1}(t)}{\overline{\beta}_n(t)}\right) \tag{38}$$

with $k_i > 0$ being positive constants.

Theorem 2: Consider the uncertain nonlinear system (1) satisfying Assumptions 1–4. Suppose that the extreme failure is detected with the supervision mechanism (13) at t_{d0}, and then, the backup actuator replaces the faulty one at t_{r0}. If the control scheme (36)–(38) is applied, the following objectives are achieved.

1) For $t \in [t_{r0}, t_1)$, the closed-loop system is stable and all the internal signals are uniformly bounded.
2) The errors recover to the prescribed performance within the prescribed time T_r after the faulty actuator replacement, i.e.,

$$|z_i(t_{r0} + T_r)| < \beta_i(t_{r0} + T_r). \tag{39}$$

Proof: The proof is similar to Theorem 1, which is omitted here. □

C. Unified Control Scheme for the Entire Process

It is interesting that the proposed control scheme can be extended to all operation process $[t_k, t_{k+1}], k = 1, 2, \ldots$. In fact, by defining t_{efk}, t_{dk}, and t_{rk} as the time instants for the $(k+1)$th period of extreme failure occurrence, extreme failure detection, and actuator replacement, respectively, then, since the controller no longer affects the actuator at extreme failure phase $t \in [t_{dk}, t_{rk})$, we can simplify design $u = 0$. The control scheme for the entire process of system operation is as follows:

$$u(t, \bar{x}_n) = \begin{cases} -k_n F\left(\frac{x_n(t) - \alpha_{n-1}(t)}{\beta_n(t)}\right), & t \in [t_k, t_{dk}) \\ 0, & t \in [t_{dk}, t_{rk}) \\ -k_n F\left(\frac{x_n(t) - \alpha_{n-1}(t)}{\overline{\beta}_n(t)}\right), & t \in [t_{rk}, t_{k+1}) \end{cases} \tag{40}$$

where the time instant t_{dk} can be determined based on

$$t_{dk} = inf\{t : |z_1(t)| \geq \beta_1(t) \text{ or } |z_i(t)| \geq \beta_i, i = 2, \ldots, n\}. \tag{41}$$

Clearly, the proposed fault-tolerant control methods in (40) are applicable to the entire system operating process, ranging from the actuator health phase, the actuator moderate failure phase, and the

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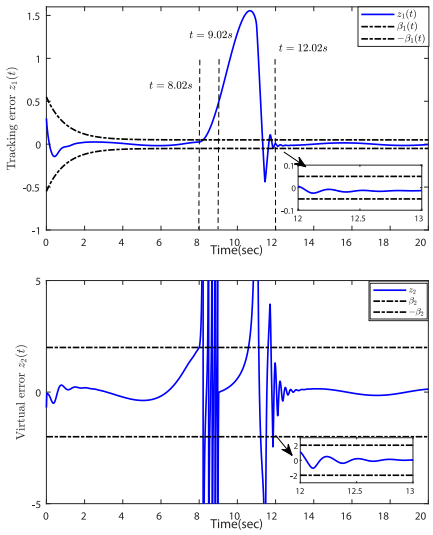


Fig. 4. Response of tracking error $z_1(t)$ and virtual error $z_2(t)$ under desired recovery time $T_r = 3$ s.

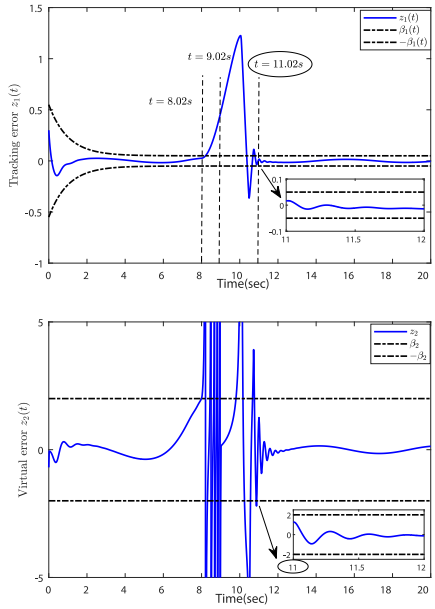


Fig. 5. Response of tracking error $z_1(t)$ and virtual error $z_2(t)$ under desired recovery time $T_r = 2$ s.

actuator extreme failure phase to the actuator post extreme failure phase. Furthermore, the control structure remains unified for all phases, even when the actuators differ before and after t_{rk}. In these unified control structure, our control objectives of prescribed performance and performance recovery are achieved concurrently. On the other hand, the extreme faults detection conditions are $|z_i(t)| \geq \beta_i$ rather than $|z_i(t)| = \beta_i$, thus some degree of detection delay is allowed in this article.

VI. Simulation Results

The longitudinal dynamics of a twin otter aircraft are used to verify the theoretical conclusions of this work. The dual-redundant actuators architecture can refer to [19, Fig. 4]. In the beginning, the actuator 1 is put into operation. If extreme failure occurs and is detected by the extreme fault supervision mechanism (13), the backup actuator 2 will be activated. The health indicator for actuator 1 is

$$\rho(t) = \begin{cases} 1 - \frac{t}{t_{ef0}}, & 0 \leq t \leq t_{ef0} \\ 0, & t > t_{ef0} \end{cases} \tag{42}$$

and the health indicator for actuator 2 is

$$\rho(t) = 1 \tag{43}$$

for all times. The studied system model is the same as those in the simulation section in [19], which can be transformed into the following form:

$$\begin{aligned} \dot{x}_3 &= x_4 \\ \dot{x}_4 &= \theta^T \phi(x) + 0.017(1 + x_1^2)u_a \\ \dot{\omega} &= \Psi_1(\chi, \omega) + \theta^T \Psi_2(\chi, \omega) \\ y &= x_3 \end{aligned} \tag{44}$$

where $\chi = [x_3, x_4]^T, \theta \in \mathbb{R}^4$, and $\phi(x) = [x_1^2 x_2, x_1^2 x_2^2, x_1^2, x_1^2 x_4]^T$. The subsystem $\dot{\omega} = \Psi_1(\chi, \omega) + \theta^T \Psi_2(\chi, \omega)$ is input-to-state stable with χ being the input and $\omega = [x_1, x_2]^T$. Thus, the dynamic of $[x_3, x_4]^T$ belongs to the form of (1). The control objective is twofold.

1) Output y tracks the trajectory $y_r = \sin(t)$ with the prescribed performance $|z_1(t)| = |y - y_r| < \beta_1(t)$ and $|z_2(t)| = |x_2 - \alpha_1| < \beta_2$ at the health and moderate actuator failure phase with $\beta_1(t) = 0.5\exp(-t) + 0.05$ and $\beta_2 = 2$;
2) System recovers from performance violation to performance satisfaction with the predefined time T_r during the post extreme failure phase.

In the simulation, the extreme failure occurrence time for actuator 1 is $t_{ef0} = 8$ s, the initial states are selected as $x(0) = [10, 0.5, 0.3, -3]^T$, the system parameters are chosen as $\theta = [0.005, 0.005, 0.005, 0.005]^T$, the control parameters are chosen as $k_1 = 2$ and $k_2 = 5$. Besides, the constraining function is selected as $F(\xi_i) = \frac{2\xi_i}{(1+\xi_i)(1-\xi_i)}$, and the deferred actuator replacement time and the recovery time are set as $T_{\text{def}} = 1$ s and $T_r = 3\,s$, respectively. As shown in Fig. 4, the errors can be ensured within the performance boundaries for $0\text{ s} \leq t < 8$ s. After the occurrence of extreme failure for actuator 1 for $t \geq 8$ s, the extreme failure can be detected at $t = 8.02$ s and the new actuator is replaced at $t = T_{\text{def}} + 8.02\text{ s} = 9.02$ s. On the other hand, after the replacement of new actuator, the errors recover to the prescribed performance within $t = T_r + 9.02\text{ s} = 12.02$ s. In addition, in order to verify that the recovery time can be predefined, we change the recovery time to $T_r = 2\,s$ and the response of errors are shown in Fig. 5. It is shown that the errors can recover to the prescribed performance boundaries within $t = 11.02$ s.

VII. Conclusion

In this article, a unified performance-based fault-tolerant control framework is proposed for multiple types/phases of actuator failure. To avoid the unnecessary actuator replacement in dynamic redundancy mechanism, the performance-based fault detector is only triggered during extreme failure occurrence phase. With the aid of error-dependent constraining function, our proposed control schemes are not only able to ensure performance satisfaction during moderate failure phase, but also able to achieve performance recovery within the predefined time during the post extreme failure phase. Extending the method to account for input saturation represents an interesting topic for future study.

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